

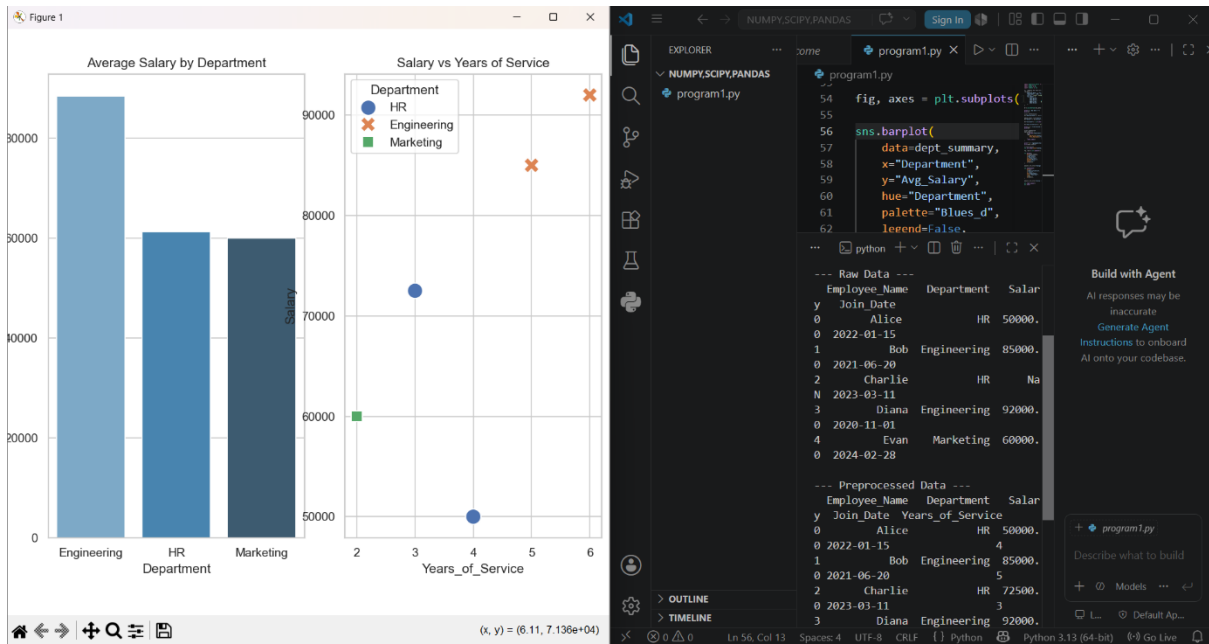
PRACTICE EXPERIMENTS

DSA0404-FDS

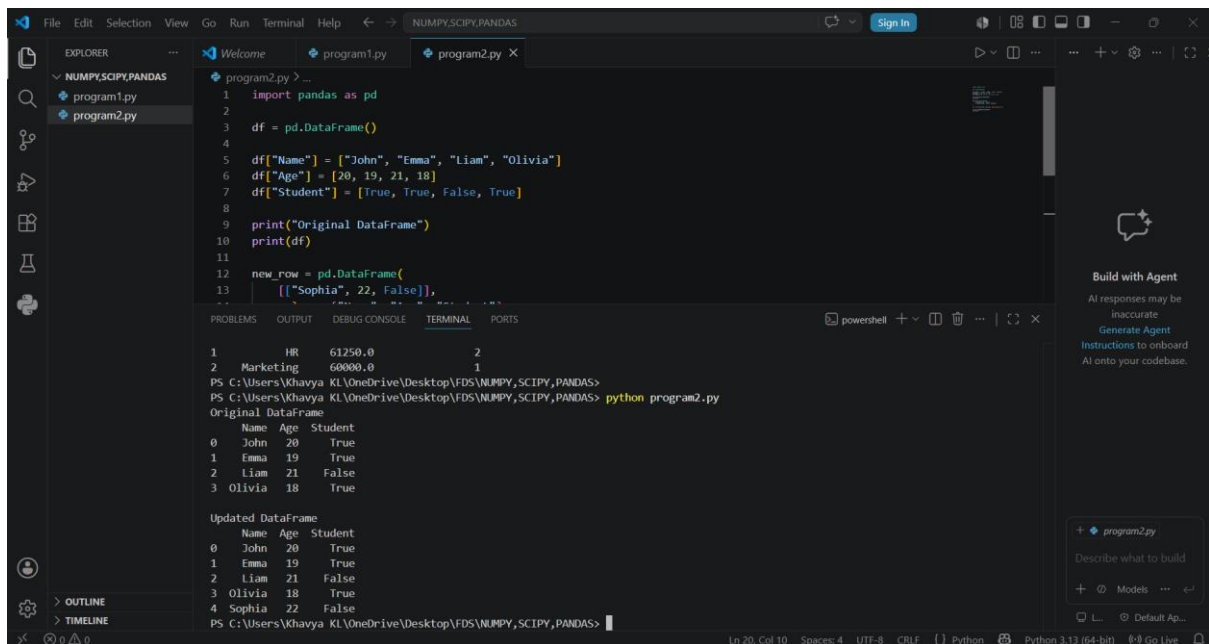
Name: K. L. Khaviya varshini

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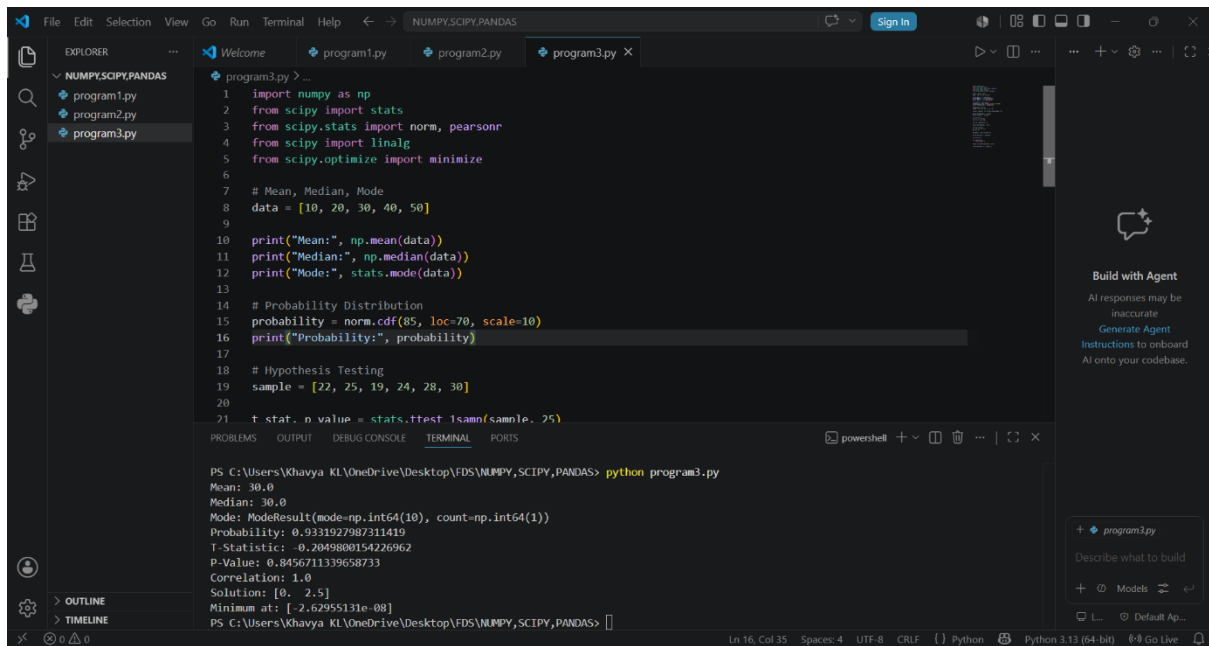
PROGRAM 1:



PROGRAM 2:

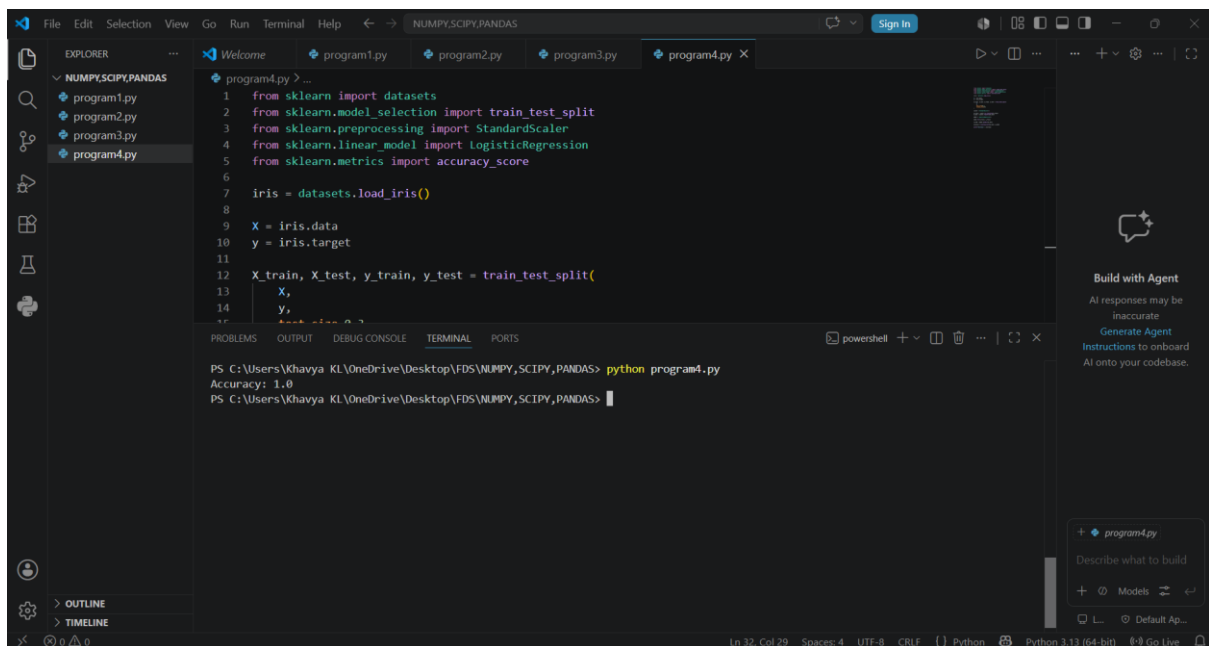


PROGRAM 3:



```
File Edit Selection View Go Run Terminal Help NUMPY, SCIPY, PANDAS
Welcome program1.py program2.py program3.py
program3.py
1 import numpy as np
2 from scipy import stats
3 from scipy.stats import norm, pearsonr
4 from scipy import linalg
5 from scipy.optimize import minimize
6
7 # Mean, Median, Mode
8 data = [10, 20, 30, 40, 50]
9
10 print("Mean:", np.mean(data))
11 print("Median:", np.median(data))
12 print("Mode:", stats.mode(data))
13
14 # Probability Distribution
15 probability = norm.cdf(85, loc=70, scale=10)
16 print("Probability:", probability)
17
18 # Hypothesis Testing
19 sample = [22, 25, 19, 24, 28, 30]
20
21 t_stat, p_value = stats.ttest_1samp(sample, 25)
22
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Khavya KL\OneDrive\Desktop\FDS\NUMPY, SCIPY, PANDAS> python program3.py
Mean: 30.0
Median: 30.0
Mode: ModeResult(mode=np.int64(10), count=np.int64(1))
Probability: 0.9331927987311419
T-Statistic: -0.2049800154226962
P-Value: 0.8456711339658733
Correlation: 1.0
Solution: [0. 2.5]
Minimum at: [-2.62955131e-08]
PS C:\Users\Khavya KL\OneDrive\Desktop\FDS\NUMPY, SCIPY, PANDAS>
```

PROGRAM 4:



```
File Edit Selection View Go Run Terminal Help NUMPY, SCIPY, PANDAS
Welcome program1.py program2.py program3.py program4.py
program4.py
1 from sklearn import datasets
2 from sklearn.model_selection import train_test_split
3 from sklearn.preprocessing import StandardScaler
4 from sklearn.linear_model import LogisticRegression
5 from sklearn.metrics import accuracy_score
6
7 iris = datasets.load_iris()
8
9 X = iris.data
10 y = iris.target
11
12 X_train, X_test, y_train, y_test = train_test_split(
13     X,
14     y,
15     test_size=0.3,
16     random_state=42)
17
18 scaler = StandardScaler()
19 X_train = scaler.fit_transform(X_train)
20 X_test = scaler.transform(X_test)
21
22 model = LogisticRegression()
23 model.fit(X_train, y_train)
24
25 y_pred = model.predict(X_test)
26 accuracy = accuracy_score(y_test, y_pred)
27
28 print("Accuracy: %.2f" % accuracy)
29
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
PS C:\Users\Khavya KL\OneDrive\Desktop\FDS\NUMPY, SCIPY, PANDAS> python program4.py
Accuracy: 1.0
PS C:\Users\Khavya KL\OneDrive\Desktop\FDS\NUMPY, SCIPY, PANDAS>
```

PROGRAM 5: